

# Simplifying ultrasound acquisition hardware using low-power coded excitation

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## Background, Motivation and Objective

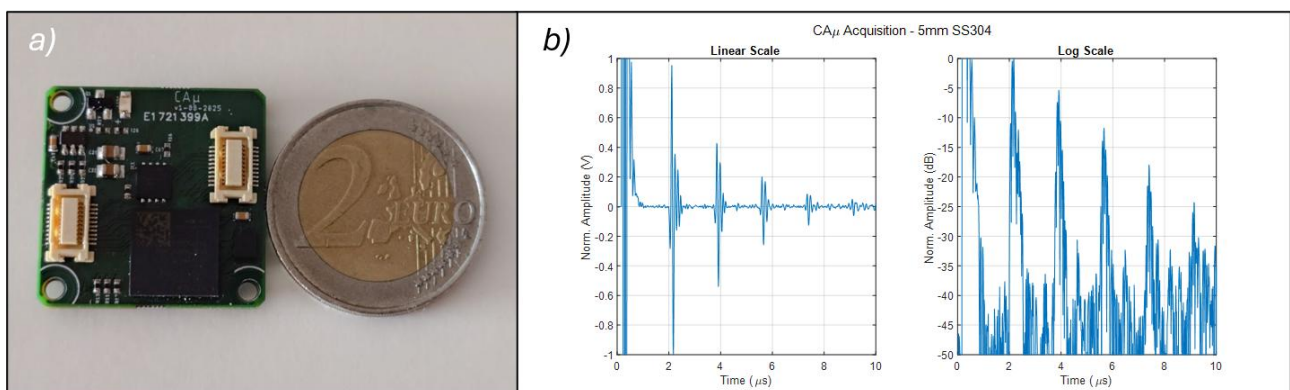
Ultrasonic signal to noise ratio (SNR) is proportional to the total transmitted energy. Conventional acquisition hardware utilises this principle and employs high amplitude, narrow excitation signals coupled with receiver amplification to produce high quality information. Whilst effective, this approach results in hardware that is complex, bulky and expensive. Coded excitation offers an alternative through temporally long, modulated transmissions, enabling more efficient hardware for IoT and wearable applications.

## Statement of Contribution/Methods

This work presents a novel ultrasonic acquisition platform built around the use of coded excitation. Specifically, TOP-CS codes are used to enable perfect in-window compression of long signals, thus enabling a high transmit energy and artifact-free information. These signals are deployed on a system capable of simultaneously transmitting and receiving ultrasound energy through a single transducer – possible via superposition and by ensuring that the transmit amplitudes sit within the receiver range. Combining this hardware with TOP-CS encoding allows for a ‘software-defined’ signal quality – the main factor to control is the signal duration.

## Results/Discussion

The system developed in this work is shown in subfigure *a*. This device uses a significantly reduced number of components compared to a conventional acquisition system, allowing it to fit within a 26x26mm footprint and reducing power consumption. Furthermore, this device uses only low-voltages – within 2Vpp – and does not require any receiver amplification. Nonetheless, the performance of this device is comparable to industry standard equipment as it supports long ( $L > 1000$ ) TOP-CS signals. Performance comparison of the system to a reference platform in an industrial application are presented – an example signal from such an application is shown in subfigure *b* in which  $>45\text{dB}$  SNR (using noise RMS) is achieved. The suitability of this device for use in IoT/wearable applications is discussed.



Subfigure *a*) the CAμ code-based acquisition system; Subfigure *b*) linear and log signals from using the CAμ to record contact pulse echo measurements of a 5mm stainless steel block with a 10MHz transducer.